

Anti-aromatase chemicals in red wine.



Estrogen synthesized in situ plays a more important role in breast cancer cell proliferation than does circulating estrogen. Aromatase is the enzyme that converts androgen to estrogen and is expressed at a higher level in breast cancer tissue than in surrounding noncancer tissue. A promising route of chemoprevention against breast cancer may be through the suppression of in situ estrogen formation using aromatase inhibitors. A diet high in fruits and vegetables may reduce the incidence of breast cancer, because they contain phytochemicals that can act as aromatase inhibitors. In our previous studies, we found that grapes and wine contain potent phytochemicals that can inhibit aromatase. We show that red wine was more effective than white wine in suppressing aromatase activity. Interestingly, our results from white wine studies suggest a weak inductive effect of alcohol on aromatase activity. On the other hand, the potent effect of anti-aromatase chemicals in red wine overcomes the weak inductive effect of alcohol in wine. Several purification procedures were performed on whole red wine to separate active aromatase inhibitors from non-active compounds. These techniques included liquid-liquid extraction, silica gel chromatography, various solid phase extraction (SPE) columns, and high performance liquid chromatography. An active Pinot Noir red wine SPE C18 column fraction (20% acetonitrile:water) was more effective than complete Pinot Noir wine in suppressing aromatase assay. This red wine extract was further analyzed in a transgenic mouse model in which aromatase was over-expressed in mammary tissue. Our gavaged red wine extract completely abrogated aromatase-induced hyperplasia and other neoplastic changes in mammary tissue. These results suggest that red wine or red wine extract may be a chemopreventive diet supplement for postmenopausal women who have a high risk of breast cancer. Further research is underway to purify and characterize the active compounds in red wine that are responsible for the inhibition of aromatase.